

ASSIGNMENT #4

ECBM E4070/E6070, Professor Aurel A. Lazar

Deadline: 11:59PM, Friday, April 3rd, 2026

In this assignment you will replicate some of the figures from the Lecture Notes #8 and #9:

- (1) Characterizing the Active Receptor Model (cont'd), An Example: Response to **Step, Ramp and Parabola** Stimuli, Lecture #8, slide #49.
- (2) I/O Characterization of the OTP Model (cont'd), An Example: Response to **Step, Ramp and Parabola** Stimuli, Lecture #8, slide #50.
- (3) I/O Characterization of the Connors-Stevens Neuron (CS Model), Lecture #8, slide #56, 2D plot on the left.
- (4) I/O Characterization of the Hodgkin-Huxley Neuron (HH Model), Lecture #8, slide #57, 2D plot on the right. Compare the I/O characterization of HH and CS models and comment on your observation.
- (5) I/O Characterization of the OTP/BSG Cascade, An Example: Response to **Step, Ramp and Parabola** Stimuli, Lecture #9, slide #17. Simulate and compare both CS and HH models; comment on your observation.
- (6) The I/O of the OTP/BSG Cascade in Steady State, The Response of the OTP/BSG Cascade to **Constant Input** Stimuli in Steady State, Lecture #9, slide #19, 2D plot on the right. Simulate and compare both CS and HH models; comment on your observation.

Task #1: Replicate the figures mentioned above. You are given a notebook, `osns.ipynb`, to aid your implementation. Below, we give some extra information, suggestions and hints:

- Use a step size of at most 10^{-5} seconds. This value will cause the output values of your simulations to be slightly different than the ones in the text, but qualitatively your figures should look similar. Both the BSG models, in particular, will benefit from smaller step sizes.
- $[\mathbf{v}]_{ron}$ cannot be negative; set negative values of $[\mathbf{v}]_{ron}$ to zero.

- Note that various state variables like $[\mathbf{x}_1]_{ron}$, $[\mathbf{x}_2]_{ron}$ and the channel state variables for the BSGs (n , m , h , a and b for the CS model and n , m , h for the HH model) cannot be outside $[0, 1]$; make sure that your code avoids such issues, should they ever occur.
- Noise added to the Connor-Stevens model could cause your simple spike detectors to detect extra spikes. You can deal with this issue by assuming the existence of a refractory period of 1 ms; that is, upon detecting a spike, disregard all other spikes found within the next 1 ms.

You are given a notebook, `osns.ipynb`, to aid your implementation. Note that you do not need to use the code in `osns.ipynb` in your implementation, but you are asked to keep your work in a notebook. Name your notebook with the code and results as `YOURUNI_assignment4.ipynb` and include it in your submission.

Submission Instructions: Your submission must include

1. The requested `.ipynb` file.
2. a `.html` file exported from `.ipynb` file by using the File>Export Notebook As...>HTML option.

Submit your work as a `.zip` file named `assignment4.YOURUNI.zip` (replace YOURUNI with your UNI) using the **Assignments** section of ECBM E4070/E6070 on Courseworks.

Please post any questions regarding this assignment on the **ED Discussion** forum. Your questions and the answers may also benefit the other students in class. If you have any other questions, please do not hesitate to contact the CA using his email address that is available on the course website.

Submission Instructions: Your submission must include

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Please post any questions regarding this assignment on the **Ed Discussion** forum. Your questions and the answers may also benefit the other students in class. If you have any other questions, please do not hesitate to contact the CA using his email address that is available on the course website.

GOOD LUCK!